



Loops Problem Bank

Section 1: Loop Fundamentals

★ Print Numbers from 1 to N

★ Print Numbers from N to 1

Print Even Numbers up to N

Print Odd Numbers up to N

Count Numbers from A to B

Multiples of N

Reverse Counting

Skip Multiples of 3

Section 2: Summation Problems

★ Sum of First N Numbers

★ Sum of Even Numbers

★ Sum of Odd Numbers

Average of N Numbers

Sum of Multiples of 3 and 5

Product of First N Numbers

Running Total Calculator

Section 3: Digit-Based Problems

★ Count Digits

★ Sum of Digits

★ Product of Digits

★ Reverse a Number

First Digit of a Number

Last Digit of a Number

Largest Digit

Smallest Digit

Even Digit Count

Odd Digit Count

Section 4: Number Properties

★ Palindrome Number

★ Armstrong Number

★ Prime Number Check



★ Perfect Number

★ Strong Number

Harshad Number

Neon Number

Duck Number

Automorphic Number

Section 5: Number Series

★ Fibonacci Series

Arithmetic Progression

Geometric Progression

Square Series

Cube Series

Power Series

Section 6: Factor Problems

★ Factorial

★ Factors of a Number

★ Count Factors

★ Sum of Factors

★ Prime Factors

Common Factors of Two Numbers

Section 7: Range Problems

★ Prime Numbers in Range

★ Armstrong Numbers in Range

Palindrome Numbers in Range

Perfect Numbers in Range

Strong Numbers in Range

Even Numbers in Range

Section 8: GCD and LCM

★ GCD of Two Numbers

★ LCM of Two Numbers

Co-Prime Check

HCF using Factors



Section 9: Pattern Fundamentals

- ★ Square Pattern
- ★ Right Triangle Pattern
- ★ Inverted Triangle Pattern
- ★ Number Triangle Pattern
- ★ Floyd's Triangle
- Alphabet Triangle

Section 10: Intermediate Patterns

- ★ Pyramid Pattern
- ★ Inverted Pyramid Pattern
- ★ Diamond Pattern
- Hollow Square
- Hollow Rectangle
- X Pattern
- Butterfly Pattern

Section 11: Real World Loop Applications

- ★ Multiplication Table Generator
- ★ ATM Transaction History Simulation
- ★ Student Marks Analyzer
- ★ Electricity Bill Summary for N Customers
- ★ Employee Salary Report
- Menu Driven Calculator